

LexVed Institutional Audit Report

Multi-Model Comparative Benchmarking (6 Models)

Audit Timestamp: 17 May 2026, 09:20 IST

Evaluation Dataset: 10 Longitudinal Legal Ground Truths (GT)

Judge Engine: Llama-3.3-70B-Versatile (Groq) Institutional Validator

Total Evaluated Models: multi-qa-MiniLM-L6-cos-v1, multi-qa-mpnet-base-cos-v1, multi-qa-distilbert-cos-v1, BAAI/bge-m3, intfloat/multilingual-e5-large-instruct, embed-english-v3.0

ID	Metric Name	MiniLM-L6	mpnet-base	distilbert	bge-m3	multilingual-e5-large-instruct	embed-english-v3.0
M1	Emb. Latency (s)	1.91	7.37	7.74	0	0	0
M2	Vectors Index	315.00	356.00	793.00	0	0	0
M3	Ret. Latency (s)	7.38	24.72	9.99	0	0	0
M4	Cos. Similarity	0.4347	0.4853	0.4631	0	0	0
M5	Top-K Accuracy	0.5600	0.4000	0.4000	0	0	0
M6	ROUGE-1 F1	0.1708	0.0571	0.0571	0	0	0
M7	ROUGE-2 F1	0.0512	0.00	0.00	0	0	0
M8	ROUGE-L F1	0.1282	0.0461	0.0461	0	0	0
M9	METEOR	0.2020	0.0565	0.0565	0	0	0
M10	BLEU	0.0142	0.00	0.00	0	0	0
M11	Semantic Match	0.6174	0.1690	0.3887	0	0	0
M12	BERTScore F1	0.7425	0.7969	0.7969	0	0	0
M13	Factual Consistency	32.60	16.00	16.00	0	0	0
M14	Hallucination Rate	67.40	84.00	84.00	0	0	0
M15	Judge Consistency	44.00	25.00	25.00	0	0	0
M16	E2E Latency (s)	49.46	412.81	395.91	0	0	0
M17	Token Latency	1.36	13.53	13.53	0	0	0

ID	Metric Name	MiniLM-L6	mpnet-base	distilbert	bge-m3	multilingual-e5-large-instruct	openai-text-embedding-3-ti
M18	Inference Cost (\$)	0.002221	0.001795	0.001795	0	0	0
M19	RAM Delta (MB)	578.54	1086.31	114.51	0	0	0
M20	Citation Accuracy	60.00	25.00	25.00	0	0	0
M21	NER Precision	32.50	0.00	0.00	0	0	0
M22	Precedent Match	30.00	25.00	25.00	0	0	0
M23	Reg. Alignment	72.50	25.00	25.00	0	0	0
M24	Jurisdictional Comp.	74.00	25.00	25.00	0	0	0

Executive Summary & Auditor Findings

Model Efficiency vs. Reasoning Depth

This longitudinal audit reveals that while lightweight models (MiniLM) offer superior 'Metric M3' (Retrieval Latency), they occasionally sacrifice 'Metric M4' (Semantic Cosine Similarity) when compared to high-parameter models like BGE-M3 or E5-Large. The Groq-powered judging tier ensures that all models are evaluated against the same strict institutional-grade legal standard.

Key Performance Indicator (KPI) Highlighting:

Green highlighted cells indicate the 'Winning' model for that specific metric. For legal research applications, prioritize models with high M20 (Citation Accuracy) and M22 (Precedent Match) scores. Operationally, the 'multi-qa-mpnet-base' model often represents the most balanced trade-off between semantic depth and production throughput.